ELECTRICAL / AUTOMATION / SYSTEMS

INFRA RESILIENCE

03

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

Can dependency aware diagnosis identify the origin of a multilayer industrial failure more accurately than threshold alarms alone?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A critical infrastructure fault diagnosis laboratory designed to distinguish initiating faults from cascades of alarms across sensors, drives, control logic and software layers.

- The interesting problem is not just alarm detection; it is separating cause from symptom.
- Best aligned with your baggage / automation experience when scoped as a conveyor or material handling system.
- A reusable benchmark suite would be more valuable than a polished dashboard alone.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Process model** → 2 STAGE **Telemetry & logs** → 3 STAGE **Causal graph** → 4 STAGE **Engineer review**

- Discrete event plant and controller simulation with timestamped state transitions and replayable event traces.
- Dependency graph or probabilistic ranking engine linking symptoms to candidate causes with supporting evidence.
- Benchmark harness that records the injected truth, available observations and the resulting ranked explanations.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ A simulated plant cannot validate certified industrial protection or live operational procedures.

⚠ Correlation is not, by itself, a causal explanation.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

INFRA RESILIENCE / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

03

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EXPERIMENTAL METHODOLOGY

01 Model a plant with drives, occupancy sensors, PLC style logic, alarms and controller states.

02 Inject single and multiple faults across motors, sensors, communications and software timing.

03 Compare threshold only methods with dependency aware ranking and quantify ambiguity under missing data.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Top 1 / top 3 initiating fault accuracy
- Detection latency and false alarm rate
- Performance under simultaneous faults and missing telemetry

REQUIRED REPOSITORY ARTEFACTS

- Fault scenario benchmark dataset
- Invariant tests for plant and controller logic
- Root cause evaluation report and replay library

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.